ResuMetrics Workflow Architecture

End-to-End Data Lifecycle, AI Pipeline & Cloud Infrastructure

System Data Flow Overview

This architectural guide traces the complete operational lifecycle of a job screening request through ResuMetrics—from client-side authentication and file ingestion, across parallelized NLP extraction, hybrid AI matchmaking, relational database persistence, to actionable recruiter workflows and cloud edge serving.

Phase 1: Ingestion & Session Orchestration
Users authenticate via registered credentials or 1-Click Instant Demo Access (Recruiter or Applicant). The client securely posts candidate resumes (PDF/DOCX) alongside job requirements. Flask streams file payloads directly into memory without disk bottlenecks.

Phase 2: Multi-Threaded Text Extraction & Cleaning
A ThreadPoolExecutor processes files concurrently. PyPDF and python-docx extract raw document strings. A single-pass regex normalization engine standardizes technology phrases ('C++' -> 'cpp', 'Node.js' -> 'nodejs'), purges formatting artifacts, and strips stop words.

Phase 3: Information Extraction & Timeline Auditing
Pattern matchers identify contact parameters (emails, phone numbers), academic degrees, institutions, and quantifiable business achievements. A timeline verification algorithm cross-checks self-reported years of experience against actual employment dates to detect timeline fraud.

Phase 4: Hybrid AI Matchmaking & RRF Scoring
The extracted text is evaluated across three decoupled scoring dimensions: (1) TF-IDF vectorization with cosine similarity, (2) Exact skill n-gram extraction, and (3) Dense semantic sentence embeddings. Reciprocal Rank Fusion (RRF) harmonizes all metrics into a normalized 0-100% match index.

Phase 5: Relational Persistence (MySQL & Hybrid Fallback)
Extracted profiles, user sessions, and comprehensive analysis history payloads are committed directly to the user's MySQL database (`resumetrics` on localhost:3306). When operating in serverless environments, the self-healing SQLiteConnectionProxy maintains uninterrupted data persistence.

Phase 6: Interactive Recruiter & Applicant Dashboards
Results render immediately in the responsive UI. Recruiters view sorted candidate cards, comparative radar charts, and skill badges. Applicants receive targeted keyword insights, missing skill roadmaps, and resume optimization advice.

Phase 7: Actionable ATS Pipeline & Automated Outreach
Recruiters move applicants across a dynamic Kanban board (Applied, Shortlisted, Interviewing, Rejected), trigger bulk shortlist notification emails via SMTP, and export executive PDF summaries or CSV spreadsheets.

Underlying Infrastructure & Technology Stack

Flask & WSGI Middleware: Powers backend routing, session management, and Vercel serverless integration via ProxyFix and VercelRoutingMiddleware.

Hybrid Database Layer: Primary enterprise MySQL storage backed by an adaptive SQLite proxy for high availability across environments.

Machine Learning Stack: Scikit-learn, SentenceTransformers, PyTorch, and NumPy executing vectorized similarity computations.

Frontend & Visualizations: Modern CSS design system, responsive UI components, Chart.js radar analytics, and real-time DOM updates.